FoodLink

User Manual

Team Green — CS 411W

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1. Introduction

FoodLink is a web-based platform designed to connect food banks across Virginia into a unified inventory network. Food banks currently operate in isolation, managing their inventory locally and relying on informal communication to coordinate resource sharing. This results in surplus food going to waste at some locations while others face critical shortages.

FoodLink solves this by providing a shared real-time inventory system, automatic surplus and shortage alerts, and a streamlined transfer request and approval workflow. Each food bank sees only their own inventory and manages their own staff accounts, while being able to view and request items from other food banks in the network.

This manual covers all core features of the FoodLink prototype: signing up, logging in, managing inventory, and coordinating transfers between food banks.

2. How to Sign Up

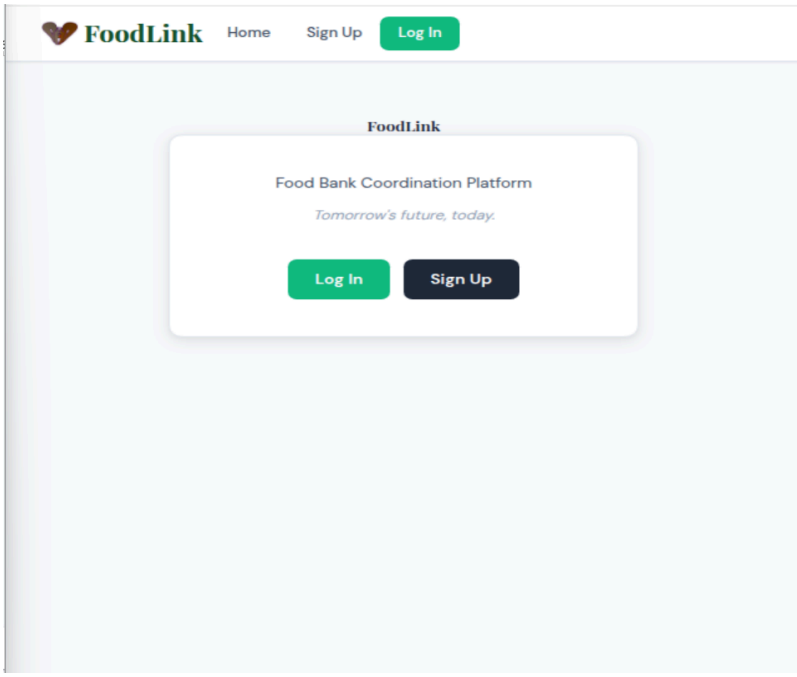
Upon accessing FoodLink for the first time, you will see the landing page with options to Log In or Sign Up. If you do not have an account, click Sign Up.

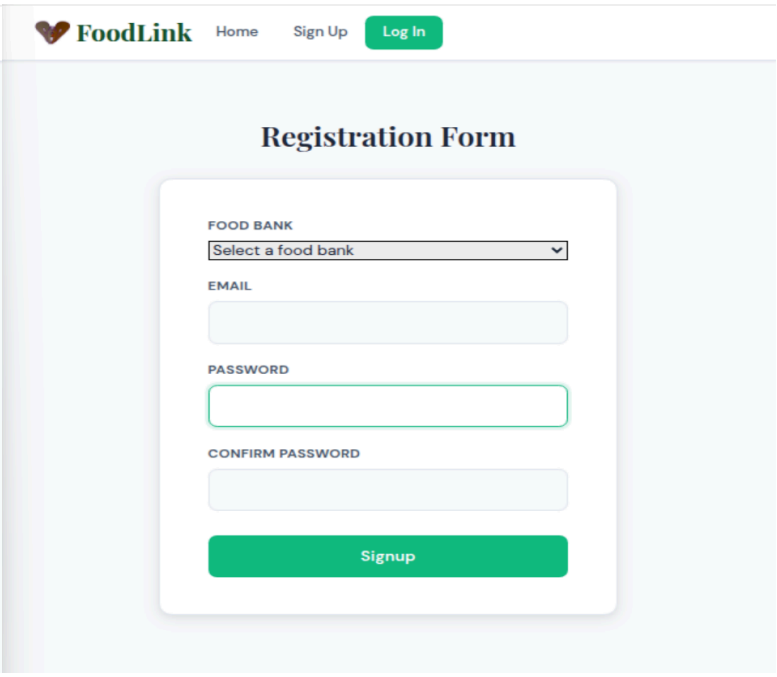
Requirements

To register you will need: a valid email address, a password meeting the security requirements (at least 8 characters, one uppercase letter, one lowercase letter, one number, and one special character), and confirmation that your food bank is listed in the system. If your food bank is not in the dropdown, contact your administrator to have it added.

Steps

1. Click Sign Up in the navigation bar.
2. Select your food bank from the dropdown list.
3. Enter your email address.
4. Enter and confirm your password.
5. Click Sign Up. You will be redirected to the login page on success.





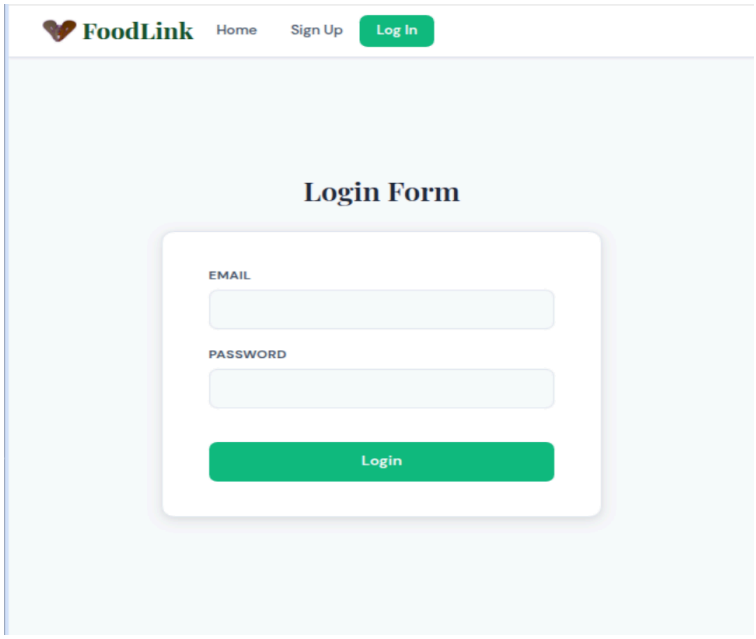
3. How to Log In

Once you have registered, you can log in to access FoodLink.

Steps

1. Click Log In in the navigation bar.
2. Enter your registered email address.
3. Enter your password.
4. Click Login. You will be redirected to the home page showing your food bank name.

If your credentials are incorrect, an error message will appear beneath the form. Check your email and password and try again.



4. How to Navigate

After logging in, the navigation bar at the top of the page gives you access to all sections of FoodLink.

Navigation Bar

Home — Returns you to the welcome page showing your food bank name and email.

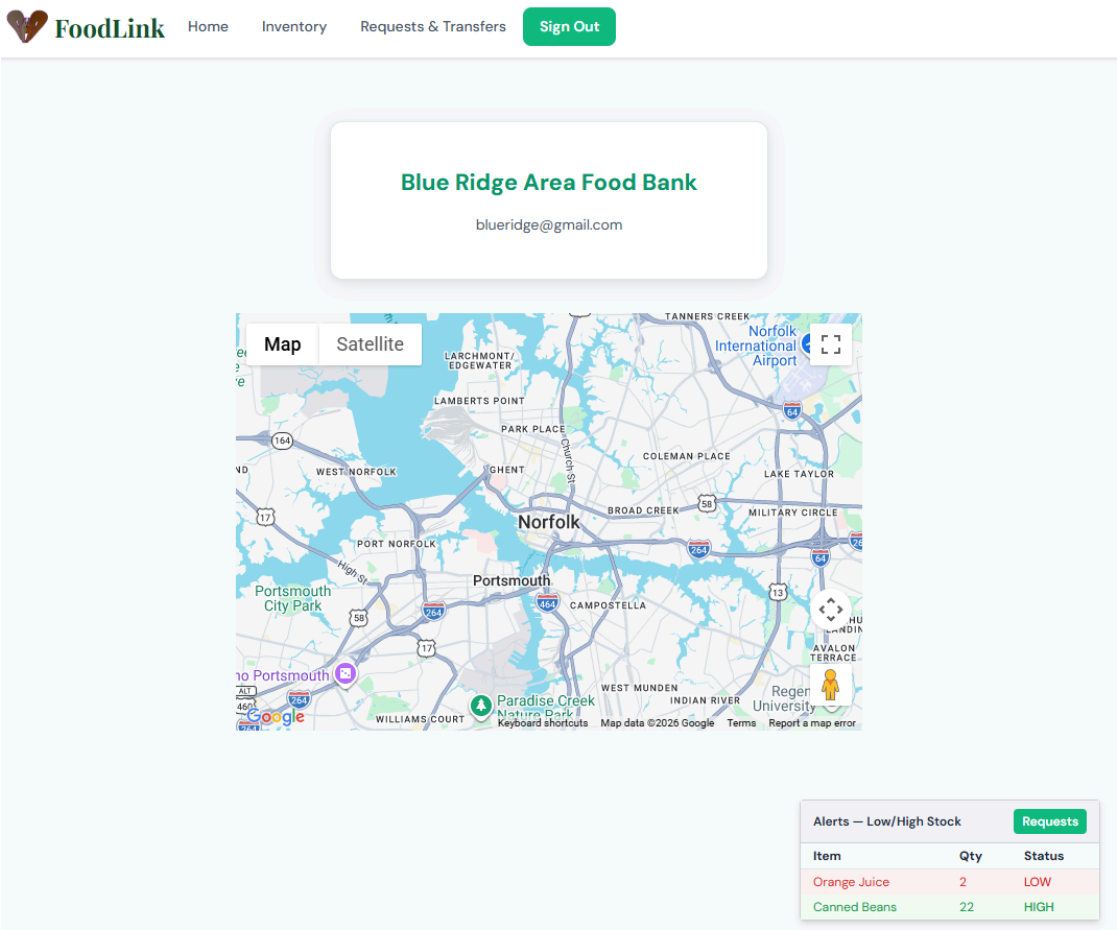
Inventory — View and manage your food bank's current inventory.

Requests & Transfers — Submit and manage transfer requests between food banks.

Sign Out — Logs you out of your account.

Alert Widget

When logged in, a small alert panel appears in the bottom right corner of every page. This widget automatically shows any inventory items that are critically low (3 units or fewer, shown in red) or overstocked (20 units or more, shown in green). Click the panel header to collapse or expand it. The Requests button in the widget navigates directly to the Requests & Transfers page.



5. How to Manage Inventory

The Inventory page displays all items currently tracked for your food bank. Each row shows the item name, category, unit of measurement, and current quantity. Rows are color coded: red for low stock (3 or fewer units), green for high stock (20 or more units), and white for normal stock.

Adding an Item

1. At the top of the Inventory page, select an item from the Item dropdown.
2. Enter the quantity in the Units field.
3. Click Add Item. The item will appear in the table below.

Updating a Quantity

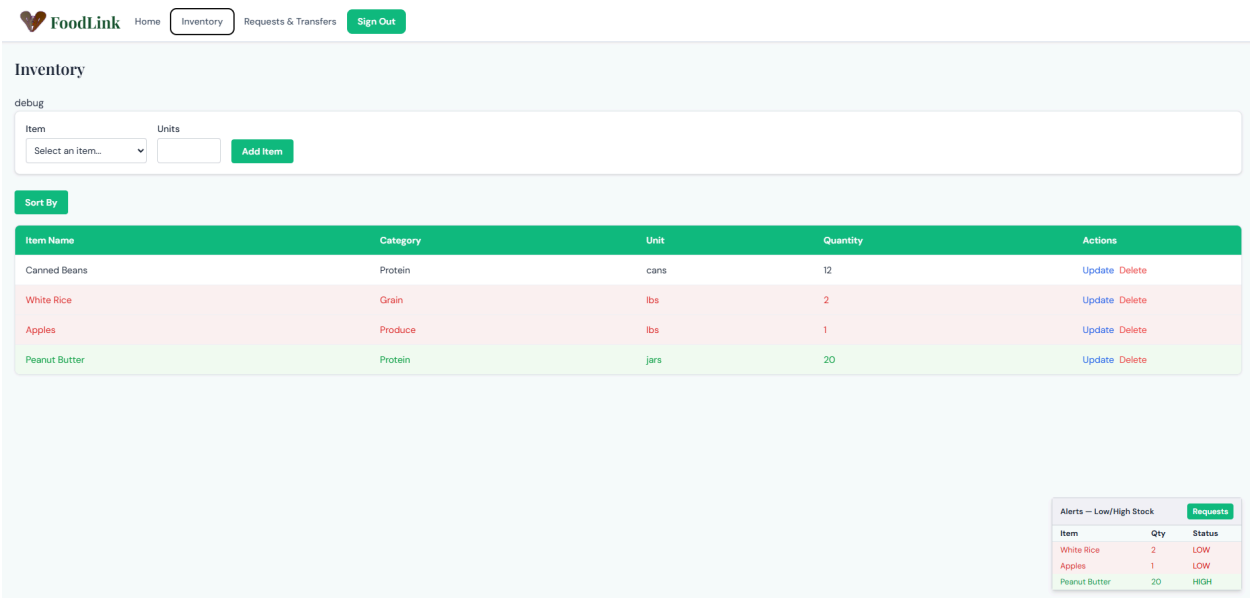
4. Find the item in the table and click Update.
5. Enter the new quantity in the input field that appears.
6. Click Save to confirm or Cancel to discard.

Deleting an Item

7. Find the item in the table and click Delete.
8. The item will be removed immediately.

Sorting

Click the Sort By button to sort the inventory table by item name or by quantity.



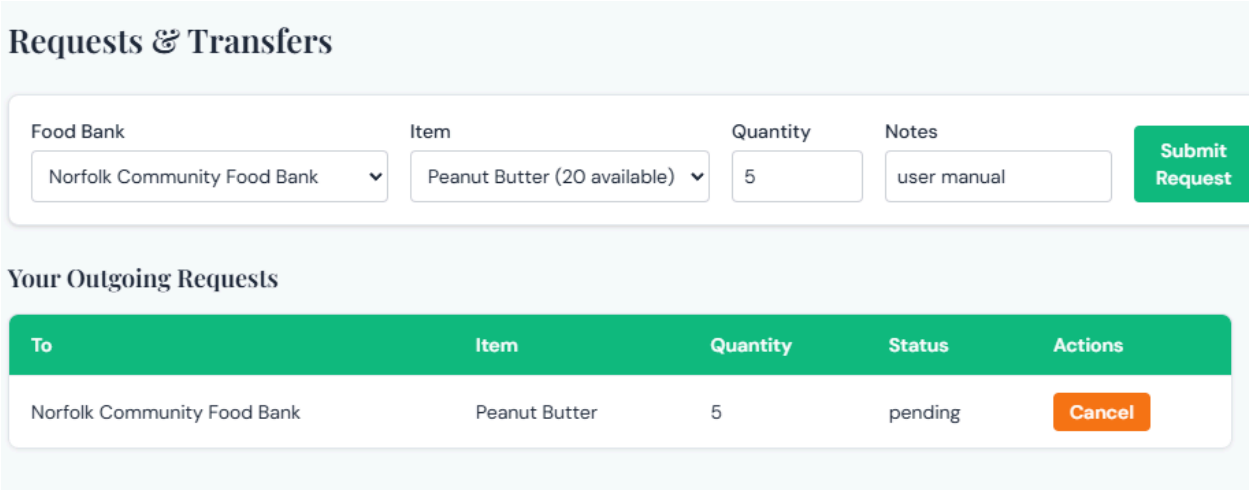
6. How to Submit a Transfer Request

The Requests & Transfers page lets you request inventory items from other food banks in the network.

Steps

1. Click Requests & Transfers in the navigation bar.
2. Select the food bank you want to request items from in the Food Bank dropdown.
3. Select the item you need from that food bank's inventory in the Item dropdown.
4. Enter the quantity you are requesting.
5. Optionally add a note explaining the request.
6. Click Submit Request. The other food bank will receive your request immediately.

Your submitted request will appear in the Your Outgoing Requests table. You can cancel it from there if needed.



7. How to Approve or Decline a Transfer

When another food bank submits a transfer request to you, it will appear automatically in the Incoming Requests table on the Requests & Transfers page.

Approving a Request

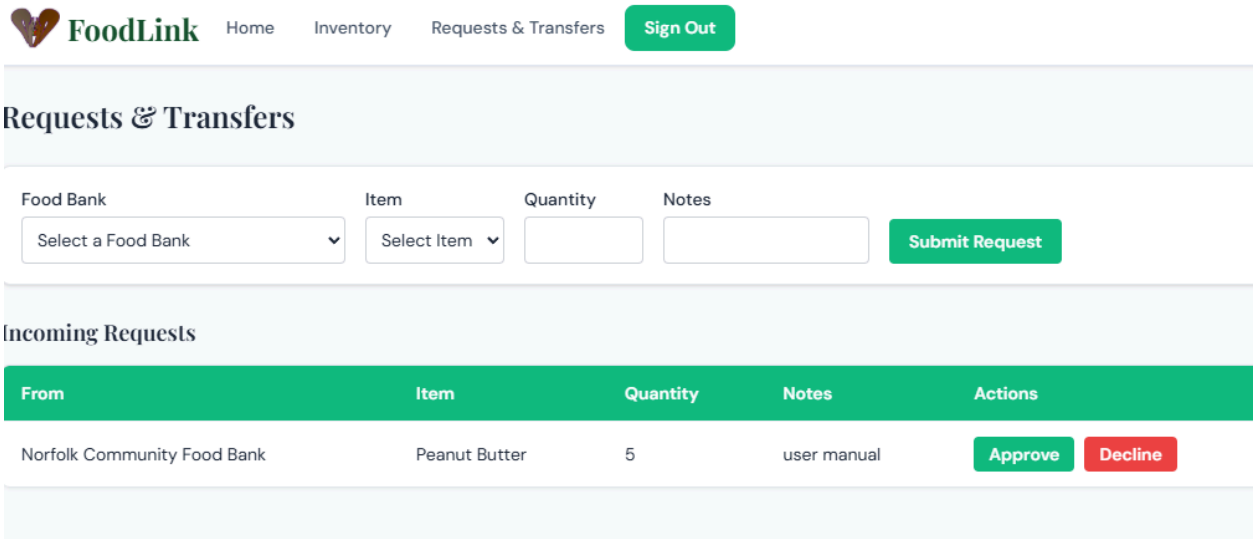
1. Go to Requests & Transfers.
2. Find the request in the Incoming Requests table.
3. Click Approve. The request status will update to approved.

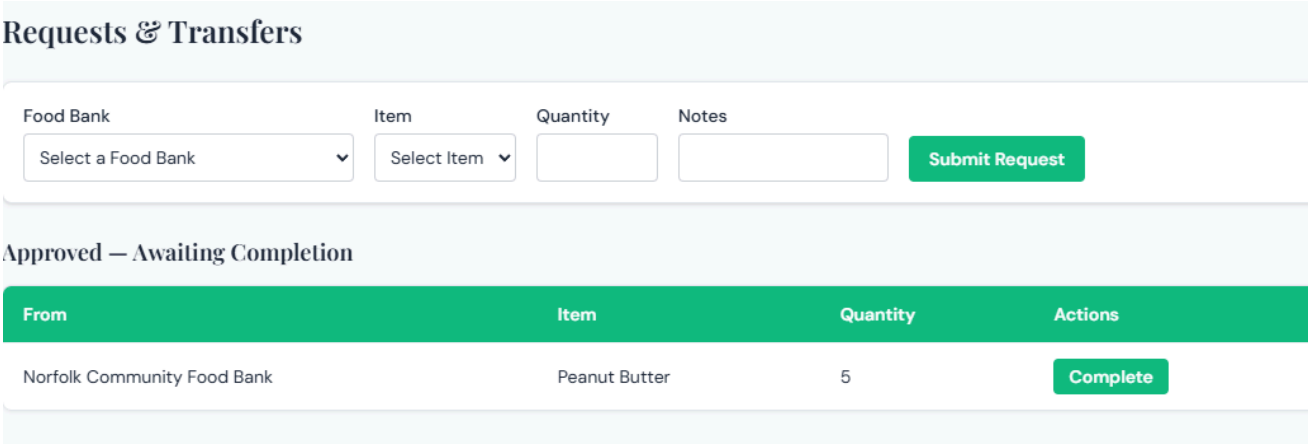
Declining a Request

4. Find the request in the Incoming Requests table.
5. Click Decline. The request will be marked as rejected.

Completing a Transfer

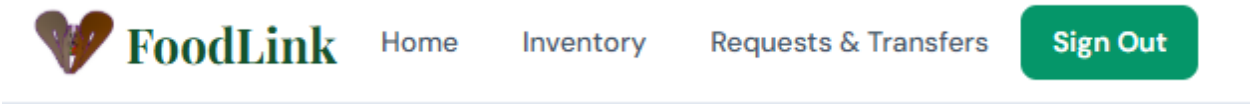
Once a request you submitted has been approved by the other food bank, it will appear in the Approved — Awaiting Completion section. Click Complete to finalize the transfer.





8. How to Sign Out

To sign out of FoodLink, click the Sign Out button in the top right of the navigation bar. You will be redirected to the login page. Your session will be cleared and you will need to log in again to access the platform.



9. Glossary & Definitions

Database — An organized collection of information that can be easily accessed, managed, and updated electronically.

Python — A high-level, interpreted programming language known for its emphasis on code readability.

PostgreSQL — A powerful, open-source, object-relational database management system used by FoodLink to store all data.

VSCode — An open-source, cross-platform source code editor developed by Microsoft.

Git — An open-source, distributed version control system used by developers to track changes in their code.

React.js — A free and open-source front-end JavaScript library used to build FoodLink's user interface.

FastAPI — A high-performance Python web framework used to build FoodLink's backend API.

API — Application Programming Interface — the layer between the frontend and the database that handles requests and responses.

Inventory — The list of food items tracked by a food bank, including quantity, category, and unit.

Transfer Request — A formal request from one food bank asking another to share inventory items.

Surplus — A quantity of an item that exceeds the high stock threshold (20 or more units), indicating more than needed.

Shortage — A quantity of an item at or below the low stock threshold (3 or fewer units), indicating a critical need.

User — A food bank staff member with a registered FoodLink account.

SRS — Software Requirements Specification — a document that describes the intended behavior and requirements of the system.